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ENBC 332

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Homework 6 Discussion and Outputs

Problem 1:

1. The null hypothesis is that the diet program has no effect on weight loss while the alternative hypothesis suggests that the diet program is effective and results in weight loss.
2. The p-value calculated, which is approximately 1.617674e-13, represents the probability of observing the sample results if the diet program was not effective or proving the null hypothesis. As a result, it shows that the diet program does give participants the opportunity to lose weight.
3. The null hypothesis is rejected because the calculated p-value is less than the significance level which is 0.05 or 5%. Alternatively, rejecting the null hypothesis suggests that the diet program is effective at lowering participants' weight.

Problem 2:

1. The null hypothesis is that the new drug causes patients' blood pressure to stay the same or be greater. The change also would cause the mean to be greater than or equal to the current mean for blood pressure. The alternative hypothesis would be that the new drug is effective at lowering blood pressure. The same also applies for the mean to be lower than the current blood pressure mean as well.
2. The p-value calculated, 7.946182e-14, is the probability that the null hypothesis is true. However, this number is very small and indicates that it is very unlikely for a treatment group to experience higher blood pressures compared to the control group randomly.
3. Since the p-value calculated is less than the significance level of 0.05 or 5%, the null hypothesis is rejected. As a result, the p-value calculated suggests that the new drug is effective at lowering systolic blood pressure.

```
> # Variables
> before <- c(78, 85, 92, 68, 74, 88, 81, 76, 95, 70,
+             79, 82, 91, 72, 80, 83, 77, 89, 84, 73)
> after <- c(75, 81, 89, 66, 71, 84, 78, 73, 90, 68,
+            76, 79, 87, 70, 77, 80, 74, 86, 80, 71)
> t_test_result <- t.test(before, after, paired = TRUE, alternative = "greater")
> t_score <- t_test_result$statistic
> p_value <- t_test_result$p.value
> cat("Results:\nt-score:", t_score, "\np-value:", p_value)
Results:
t-score: 17.59189
p-value: 1.617674e-13
> # Variables
> treatment_group <- c(-12, -8, -10, -5, -7, -13, -9, -11, -6, -10, -15, -8, -11,
+                      -14, -9, -13, -8, -12, -10, -7, -11, -9, -14, -10, -12)
> control_group <- c(-3, -4, -2, -5, -6, -3, -4, -5, -2, -1, -3, -4, -6,
+                    -3, -2, -4, -5, -2, -3, -1, -4, -3, -6, -2, -4)
> # Computations
> t_test_result <- t.test(treatment_group, control_group, alternative = "less")
> t_score <- t_test_result$statistic
> p_value <- t_test_result$p.value
> cat("Results:\nt-score:", t_score, "\np-value:", p_value)
Results:
t-score: -11.14366
p-value: 7.946182e-14
```